

Algebraic Procedure to finding Local Maximum and Minimum Points

1. Find $f'(x)$ and factor it
2. Determine all of the critical numbers by finding all numbers that make $f'(x) = 0$ or $f'(x)$ undefined.
3. Order your critical numbers from most negative to most positive and use them to create an interval chart.
4. Complete your interval chart and use it to determine if $f'(x)$ changes its sign at c . Note if it's a local min or a local max by viewing the type of sign change of $f'(x)$.
5. If a local min or local max exists at c then determine $f(c)$ to find the y coordinate of your local min or local max point.

If you are asked to determine Absolute Maximum and Minimum points in an interval $x \in [a, b]$ then

6. Evaluate $f(a)$ and $f(b)$ and compare the end points of the interval with the local max and local min points you found in steps 1-5.

Ex) Given $f(x) = 2x^3 - 3x^2 - 12x + 5$ determine

- a) The critical numbers of $f'(x)$
- b) The Local Max and Local Min points
- c) The Absolute Max and Absolute Min points in the interval $x \in [-2, 4]$
- d) Make a rough sketch of $f(x)$

a) $f'(x) = 6x^2 - 6x - 12$

$$= 6(x-2)(x+1)$$

critical #'s @ $x = 2, -1$

b)

Interval	$(-\infty, -1)$	$(-1, 2)$	$(2, \infty)$
$x-2$	-	-	+
$x+1$	-	+	+
$f'(x)$	+	-	+
$f(x)$	↑	↓	↑

Local max: $(-1, 12)$

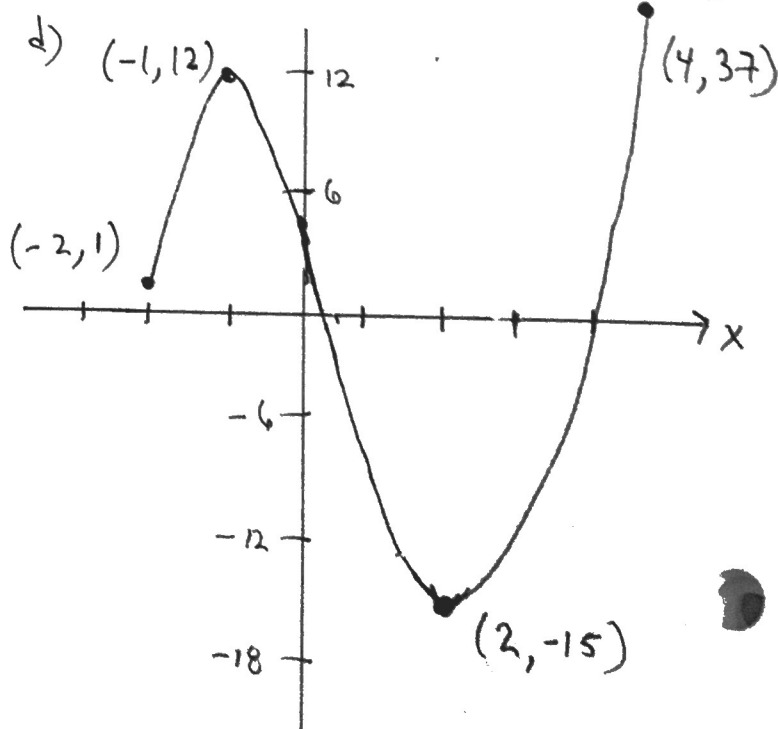
Local min: $(2, -15)$

c) $f(-2) = 1$

$$f(4) = 37$$

Absolute Min @ $(2, -15)$

Absolute Max @ $(4, 37)$



x) Given $f(x) = 4x^3 + 3x^2 - 6x + 1$ determine

a) The Local Max and Local Min points

b) Make a rough sketch of $f(x)$

a) $f'(x) = 12x^2 + 6x - 6$

$$= 6(2x-1)(x+1)$$

critical #'s: $x = \frac{1}{2}, -1$

Interval	$(-\infty, -1)$	$(-1, \frac{1}{2})$	$(\frac{1}{2}, \infty)$
$2x-1$	-	-	+
$x+1$	-	+	+
$f'(x)$	+	-	+
$f(x)$	↑	↓	↑

Max @ $x = -1$; min @ $x = \frac{1}{2}$

Ex) Given $f(x) = (1-x^2)^2 - 2$ determine

a) The Local Max and Local Min points

b) Make a rough sketch of $f(x)$

a) $f'(x) = 2(1-x^2)(-2x)$

$$= -4x(1-x^2)$$

critical #'s: $x = 0, \pm 1$

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
-4	-	-	-	-
x	-	-	+	+
$1-x$	+	+	+	-
$1+x$	-	+	+	+
$f'(x)$	-	+	-	+
$f(x)$	↓	↑	↓	↑

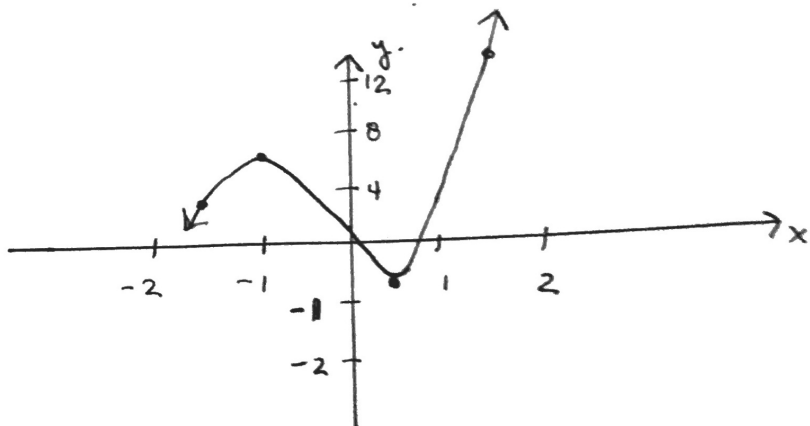
min. @ $(-1, -2)$ & $(1, -2)$

max. @ $(0, -1)$

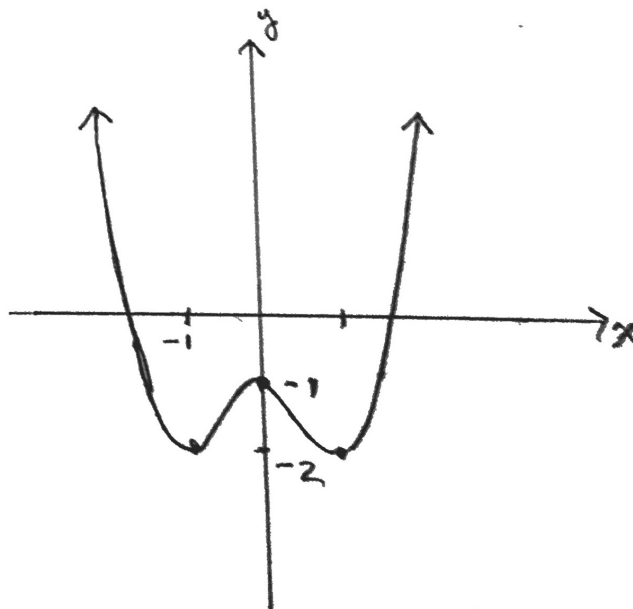
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Local max @ $(-1, 6)$

Local min @ $(\frac{1}{2}, -\frac{3}{4})$



b)



Ex) For the cubic function $f(x) = ax^3 + bx^2 + cx + d$, find the values of a , b , c , and d such that there is a local maximum at $(-2, 13)$ and a local minimum at $(0, 5)$. [$a = 2, b = 6, c = 0, d = 5$]

$$f'(x) = 3ax^2 + 2bx + c$$

Sub. $x = -2$ into $f'(x)$ and set $= 0$

$$12a - 4b + c = 0 \quad (1)$$

sub. $x = 0$ into $f'(x)$ and set $= 0$

$$\boxed{c = 0}$$

sub $(0, 5)$ into $f(x)$.

$$\boxed{d = 5}$$

sub $(-2, 13)$ into $f(x)$

$$-8a + 4b + 5 = 13$$

$$-8a + 4b = 8 \quad (2)$$

$$\textcircled{1} + \textcircled{2}$$

$$4a = 8$$

$$\boxed{a = 2}$$

sub. $a = 2$ into $\textcircled{2}$

$$\boxed{b = 6}$$

$\therefore$ Given $f(x) = \frac{x^2 - x + 1}{x^2 + 1}$ determine the Local Max and Local Min points $\left[(-1, \frac{3}{2}), (1, \frac{1}{2})\right]$

$$f'(x) = \frac{(2x - 1)(x^2 + 1) - (x^2 - x + 1)(2x)}{(x^2 + 1)^2}$$

$$f'(x) = \frac{\cancel{2x^3} + \cancel{2x} - x^2 - 1 - \cancel{2x^3} + 2x^2 - \cancel{2x}}{(x^2 + 1)^2}$$

$$f'(x) = \frac{x^2 - 1}{(x^2 + 1)^2}$$

critical #'s: $x = \pm 1$

Interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$x + 1$	-	+	+
$x - 1$	-	-	+
$(x^2 + 1)^2$	+	+	+
$f'(x)$	+	-	+
$f(x)$	$\uparrow$	$\downarrow$	$\uparrow$

$\therefore$ Local Max. $\left(-1, \frac{3}{2}\right)$

Local Min. $\left(1, \frac{1}{2}\right)$